

Research on Security Analysis and Testing Technology of Avionics Systems Based on MIL-STD-1553B

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Abstract: The avionics bus system is the core system for aircraft information processing, undertaking critical tasks of data communication and control system transmission. The security of avionics buses is a key issue to be addressed in the field of network security testing technology. This paper develops a kill chain threat model specific to MIL-STD-1553B (hereinafter referred to as 1553B) avionics systems by referencing traditional security kill chain research. It constructs a simulation environment for avionics bus systems based on RPCIE1553 boards and conducts research on security-related testing technologies using the 1553B avionics bus, which is widely applied in military and industrial fields. Since security issues in avionics bus systems may affect the normal operation of the entire aviation system, research on network security test and evaluation technologies for avionics buses is particularly important.

Keywords: Avionics Bus System; 1553B Avionics Bus; Security Analysis and Testing

I. INTRODUCTION

This paper aims to analyze network security issues in avionics bus systems and conduct security analysis and testing on such systems. For this purpose, we select the MIL-STD-1553B bus standard, which is widely used and representative in military and industrial fields, as the research object. Proposed by the U.S. military, this bus standard serves as the backbone avionics bus for military aircraft, featuring real-time performance, reliability, and security, but also contains potential security vulnerabilities, which will be thoroughly studied in this paper.

Currently, security solutions for identifying and preventing network attacks on 1553B communication buses are relatively scarce. Traditional security solutions (such as firewalls, intrusion detection, and malware detection) are not applicable to 1553B buses and require extensive adjustments and configuration changes based on the 1553B bus communication protocol [1]. Guided by the concept of "defense through offense," this paper studies the kill chain model and threat injection capabilities for avionics buses, explores in depth the network security issues of avionics bus systems, and provides theoretical support and practical guidance for the safe operation and test evaluation methods of avionics systems.

II. NETWORK SECURITY COMPLIANCE OF 1553B AVIONICS BUS SYSTEMS

A. Analysis of Network Communication Protocol Compliance Issues in 1553B Avionics Bus Systems

Problems exist in the compliance of avionics bus network communication protocol design, resulting in communication mechanisms that do not meet network security compliance requirements. Attackers may read messages transmitted over the bus and impersonate legitimate nodes to launch attacks through offensive means. According to literature analysis and experiments, when an avionics bus system uses the 1553B bus protocol for communication, all nodes are connected to a shared communication medium, and communication lacks authentication, authorization, and encryption mechanisms. This allows each sending node to directly transmit messages to other receiving nodes [2]. Therefore, we summarize the potential security issues as follows:

(1) Lack of authentication: Remote Terminals (RTs) can participate in bus communication merely by establishing legitimate addresses and message formats, indicating that the authentication mechanism of the avionics bus system may not meet security requirements.

(2) Lack of authorization: Without access control mechanisms, attackers can disguise Bus Controllers (BCs) to gain system privileges, execute arbitrary functions, and access all devices, meaning the authorization mechanism of the avionics bus system may not meet security requirements.

(3) Lack of encryption: Addresses, commands, and data are transmitted in plaintext, which can be easily captured and parsed, enabling attackers to understand the current system status and control methods. Thus, the encryption mechanism of the avionics bus system may not meet security requirements.

B. Exploitation of Vulnerabilities Caused by Compliance Issue

With the iteration of technologies, vulnerability issues faced by avionics bus systems have become increasingly prominent. These vulnerabilities are influenced by compliance issues in avionics bus communication, i.e., the lack of authentication, authorization, and encryption in network communication makes vulnerability exploitation relatively unimpeded.

Ultimately, this may lead to interception, tampering, or forgery of avionics system communication messages.

Vulnerability exploitation methods caused by compliance issues include bus network scanning and detection, eavesdropping trojans, bus virus attacks, impersonation of legitimate bus nodes, and bus privilege acquisition. These attack methods are the main threats that avionics bus systems may face. A brief introduction to these exploitation methods is provided below, with detailed descriptions in the next chapter on vulnerability testing methods:

- (1) Bus network scanning and detection: Attackers exploit the vulnerability mechanisms of the system (lack of authentication, authorization, and encryption) to broadcast and detect active assets and idle addresses in the bus network.
- (2) Eavesdropping trojans: Eavesdropping trojans forge or deceive assets in the bus to obtain bus communication messages and lurk in the bus to discover more vulnerabilities.
- (3) Bus virus attacks: Viruses can overload the bus system by sending a large number of invalid requests or junk data, or maliciously construct valid program branches to attack the bus, ultimately disrupting or controlling bus access services. This typically manifests as severe occupation of address resources or registers, program crashes, and backdoor access.
- (4) Impersonation of legitimate bus nodes: Attackers forge or impersonate legitimate nodes to communicate in the bus, seize session privileges, and execute high-risk commands. This can lead to critical session interference, message tampering, man-in-the-middle attacks, etc., ultimately disrupting the normal operation of the system. Such attacks are often difficult

to detect because they appear to originate from legitimate nodes.

- (5) Bus privilege acquisition: Attackers take over assets in the bus through the above attack methods to ultimately gain bus privileges.

III. RESEARCH ON VULNERABILITY TESTING METHODS

This chapter introduces the track of kill chain threat modeling and specific methods of threat injection.

A. Kill Chain Threat Modeling

We argue that the distinct characteristics of avionics buses and their special management and operation requirements result in kill chains triggered by vulnerabilities that differ fundamentally from traditional internet security at the methodological level. For example, prevalent internet attack chain models and ATT&CK threat frameworks assume that host Ips in computer networks are mutually/remotely reachable, operating environments are roughly identically configured, and there is space for attack payloads to reside. However, these technical prerequisites do not apply to avionics buses, requiring us to first form a correct understanding and grasp of their attack and penetration activities at the methodological level.

To address this, we have designed an attack model potentially suitable for bus threat injection. Attack and penetration activities on avionics bus networks need to be described from at least five aspects: (1) attack methods; (2) attack chains; (3) attack harms; (4) target objects. The attack model also requires these four types of attributes for support, as shown in Figure 1 [3].

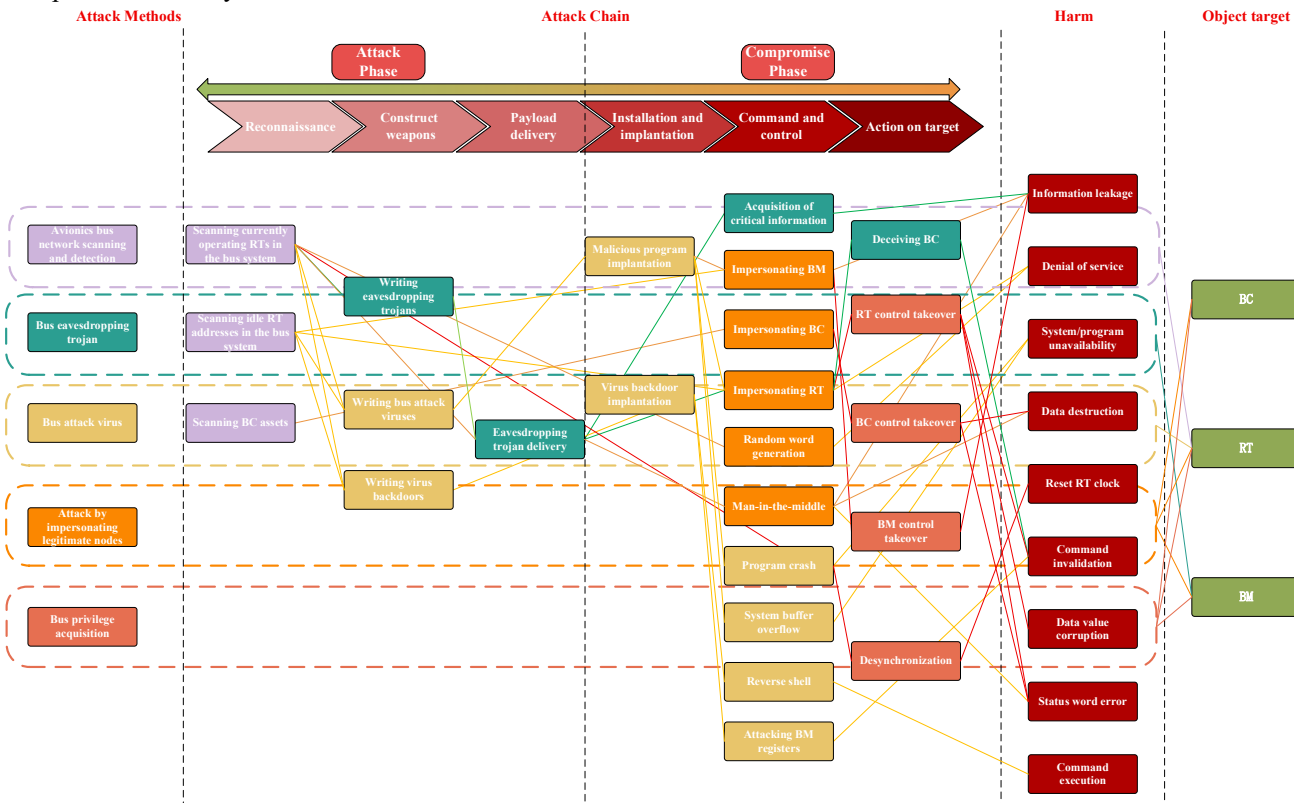


Figure 1. Kill Chain Threat Modeling for Bus Attacks

Overall, attack and penetration of avionics bus networks can be divided into two phases: (1) the attack phase; (2) the compromise phase. The attack chain framework is as follows:

(1) Target reconnaissance: Gather basic information about BCs and RTs in the avionics bus network; then construct weapons, i.e., write trojan programs for eavesdropping on the avionics bus network.

(2) Payload delivery: Implant eavesdropping trojans into the target system to obtain real-time Bus Monitor (BM) information, facilitating the exploitation of vulnerabilities in the target system and privilege acquisition in subsequent steps.

(3) Installation and implantation: Implant malicious viruses into the target system.

(4) Command and control implementation: Control the target system through malicious viruses and gain privileges. During this process, attackers can impersonate legitimate bus nodes to launch attacks in the controlled system.

(5) Action on target: Acquire bus privileges to achieve higher-level control.

B. Threat Injection Methods

The test and evaluation of avionics buses follow the above process. This section introduces detailed threat injection methods achievable through the kill chain.

(1) Avionics bus network scanning and detection

Scanning currently operating RTs in the bus system: In an avionics bus network, all devices receive the same information through bus broadcasting. RTs promptly acquire and respond to various important information to maintain system collaboration. Scanning the bus system can identify currently operating RTs.

Scanning idle RT addresses in the bus system: Idle RT addresses refer to addresses temporarily unused in the network. By scanning the bus system, occupied addresses can be identified, and unoccupied idle addresses can be analyzed through algorithmic operations.

BC asset scanning: Scan the activation and redundancy status of BCs in the bus system.

(2) Bus eavesdropping trojans

Eavesdropping trojan delivery: Trojans with BM monitoring capabilities eavesdrop on bus communication, feature self-starting functions, and monitor avionics bus transmission content in real time.

Critical information acquisition: Trojans can obtain critical information on the bus, causing data leakage.

Deceiving BCs: Trojans can impersonate RTs to deceive BCs and invalidate BM monitoring information.

(3) Bus attack viruses

Malicious program implantation: Simulate 31 RTs occupying resources to cause denial of service.

Program crashes: Buffer overflow crashes in BC simulation programs.

Virus backdoor implantation: Disguise as RTs to access the bus, seize BC functions, and gain system privileges.

System buffer overflow: Call malicious programs to send payloads, causing blue screens in the simulation system.

Attacking BM hardware pointer registers: Seize BM register addresses to invalidate monitoring data in BM service communication.

(4) Impersonation of legitimate bus nodes

Attackers impersonate legitimate nodes, disguising themselves as part of the bus system to undermine bus stability and security. Such attacks target bus communication systems, deceiving other nodes to obtain illegal benefits or achieve malicious purposes.

Impersonating BCs: Attackers impersonate BCs to send and receive messages, controlling bus communication processes and outcomes.

Impersonating RTs: Similar to impersonating BCs, but with the role of RTs.

Impersonating BMs: Attackers impersonate BMs to send and receive messages, monitoring bus communication processes and outcomes.

Random word generation: Message conflicts on the bus, rendering legitimate messages unreadable. This indicates that multiple messages may be sent simultaneously during bus communication, causing conflicts and unreadability.

Man-in-the-middle: By impersonating RTs or BCs, attackers alter normal data and data queues in memory, tamper with messages, and disrupt normal communication, which can be detected through BM monitoring of data changes.

(5) Bus privilege acquisition

Taking over BC control to issue commands to RTs: Data tampering/destruction, status word processing, etc.

Taking over RT control to return status to BCs: Command invalidation, status word processing, data word corruption, etc.

Taking over BM control for information leakage: Leakage of command words, status words, data words.

Desynchronization: Resetting RT clock synchronization.

IV. SURVIVABILITY TEST EVALUATION PROCESS AND RESULTS

Based on the above vulnerability testing methods, we designed a survivability test evaluation process by building an environment simulating normal communication between BCs and RTs for experimental research. In this simulation, the attack is carried out by impersonating a remote terminal after initially gaining access to a node, rather than by forcing the bus controller to go offline. This method exploits the lack of an authentication mechanism in the 1553B protocol to inject malicious messages and tamper with data during transmission.

A. Preparation of Bus System Simulation Environment

Compared with traditional internet security environments, we constructed a simulation environment suitable for survivability evaluation of avionics bus systems. This

environment is built based on RPCIE1553 boards [4]. The 1553B bus studied in this paper uses half-duplex communication for data transmission. According to tests and

literature research, the bus channel connection method is shown in Figure 2 [5].

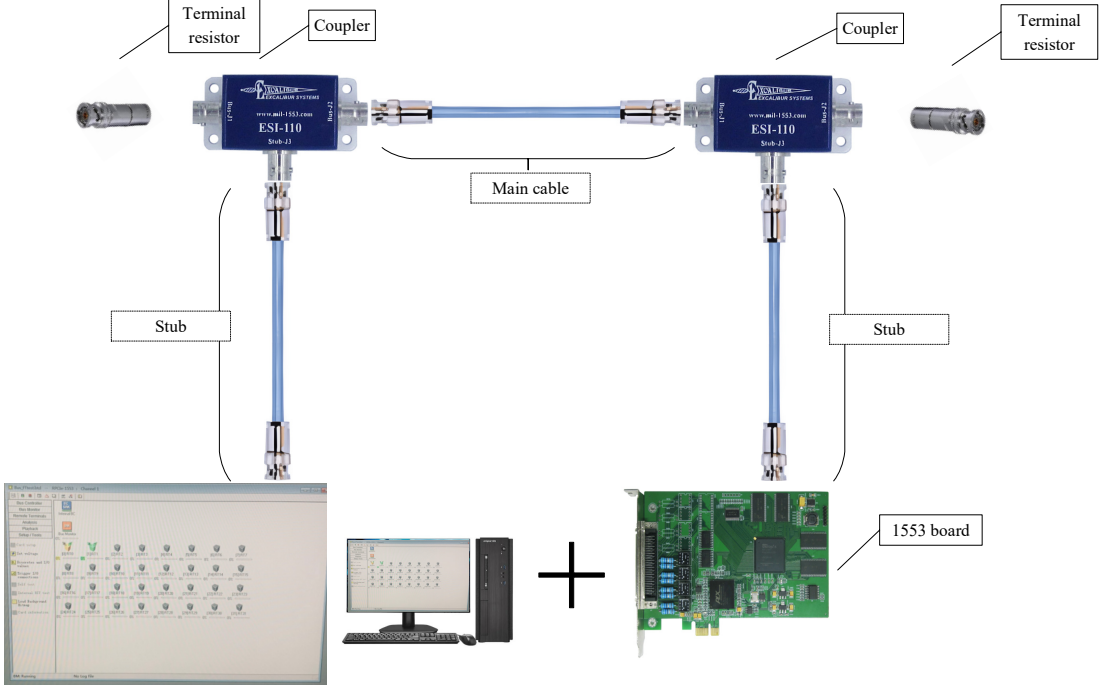


Figure 2 Wiring Diagram for Channel A or Channel B

In this experimental environment, we used industrial computers, RPCLE1553 boards, couplers, terminal resistors, and data buses (main cables and stubs) for wiring. The industrial computer is connected to the RPCLE1553 board via a PCIE interface, while couplers and terminal resistors connect the data bus to devices. Additionally, components such as BCs, RTs, and BMs can be connected separately or built in the industrial computer via simulation software. This experimental environment adopts the latter approach, with each component networked according to experimental research and literature [6].

B. Preparation of Survivability Test Tools

The survivability test tools used are self-developed programs, developed by calling 1553B bus protocol communication API interfaces for test evaluation. The design logic flow of the evaluation tool is shown in Figure 3.

C. Operation of the Simulation Environment

The simulation environment was started, and the BC channel was selected as shown in Figure 4.

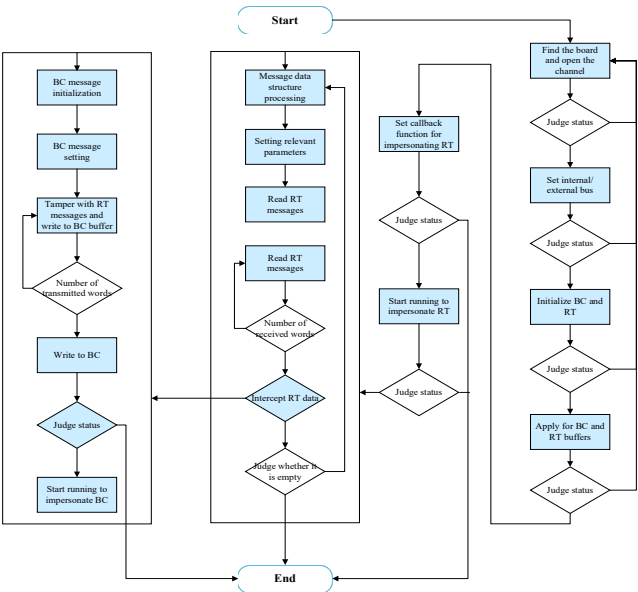


Figure 3. Program Logic Flowchart

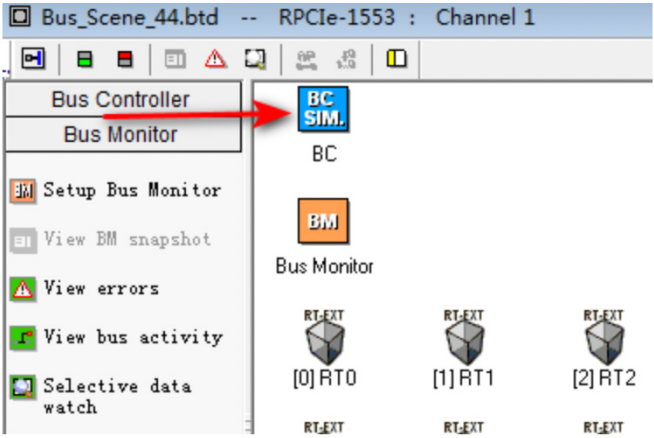


Figure 4. Simulation Channel Construction

Routine communication data was simulated as shown in Figure 5.

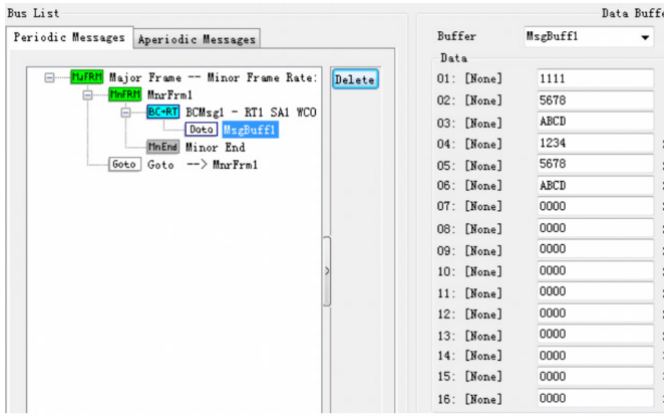


Figure 5. Message Content Presetting

The communication environment was operated as shown in Figure 6.

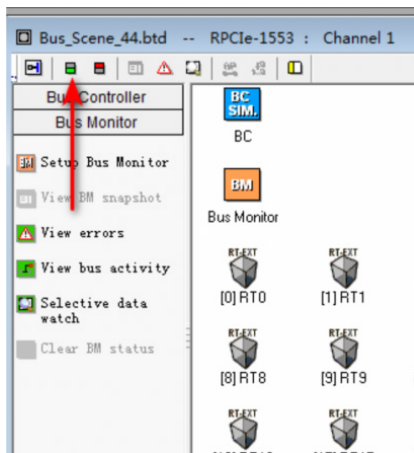


Figure 6. Running the Simulation Environment

The information monitored by BM was consistent with the simulated communication content, indicating successful operation of the simulation, as shown in Figure 7.

Hex	Label	Data	Hex
01 1111	[None]	1111	[None]
02 5678	[None]	5678	[None]
03 ABCD	[None]	ABCD	[None]
04 1234	[None]	1234	[None]
05 5678	[None]	5678	[None]
06 ABCD	[None]	ABCD	[None]
07 0000	[None]	0000	[None]
08 0000	[None]	0000	[None]
09 0000	[None]	0000	[None]
10 0000	[None]	0000	[None]
11 0000	[None]	0000	[None]
12 0000	[None]	0000	[None]
13 0000	[None]	0000	[None]
14 0000	[None]	0000	[None]
15 0000	[None]	0000	[None]

Figure 7. BM Monitoring Status

D. Operation of Test Tools

A man-in-the-middle attack was launched using the survivability test tool, causing data to increment cyclically and destroying the original data, as shown in Figure 8.

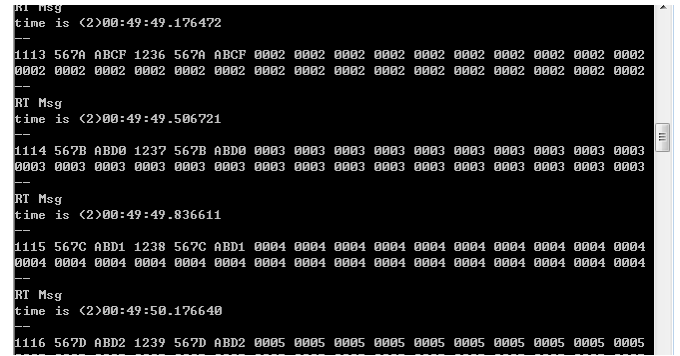


Figure 8. Man-in-the-Middle Attack on Bus Communication Caused by Test Tool Operation

E. Test Results

The test was valid, and messages monitored by BM changed with the attack. We successfully tampered with BC data by impersonating legitimate nodes, took over BC privileges, and caused harms such as destruction of normal data, as shown in Figure 9.

Hex	Label	Data	Hex
01 1116	[None]	1116	[None]
02 567D	[None]	567D	[None]
03 ABD2	[None]	ABD2	[None]
04 1239	[None]	1239	[None]
05 567D	[None]	567D	[None]
06 ABD2	[None]	ABD2	[None]
07 0005	[None]	0005	[None]
08 0005	[None]	0005	[None]
09 0005	[None]	0005	[None]
10 0005	[None]	0005	[None]
11 0005	[None]	0005	[None]
12 0005	[None]	0005	[None]

Figure 9. Abnormal BM Monitoring Status

V. CONCLUSIONS AND OUTLOOK

Through the construction of an experimental environment, this paper conducted in-depth analysis of the network security compliance of avionics bus systems, revealing potential security risks in the 1553B communication protocol. These risks may lead to vulnerability exploitation, and research on vulnerability testing methods was carried out. Finally, the existence of potential risks was confirmed through survivability test results. The methods and results presented in this paper provide a technical foundation for subsequent security evaluation of avionics bus networks in aerospace vehicles.

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